

Identity and Persistence: Structural Continuity in Dimensional Human Field Theory (DHFT)

DHFT Technical Note: Identity and Persistence

Cara Tonucci

Independent Researcher

April 14, 2026 — Version 1

Cara.Tonucci@gmail.com

ORCID: <https://orcid.org/0009-0003-9048-5821>

Abstract

This document defines identity and persistence as structural properties within Dimensional Human Field Theory (DHFT). The system is specified over invariant variables Load (L), Capacity (C), and Boundary (B).

Identity is defined as the persistence of a system across time under varying load conditions. The relations defined here establish continuity independent of momentary system state.

No additional variables are introduced. Identity is treated as a property of system continuity within the constraints of DHFT.

I. Scope

This document defines identity as a structural property of system persistence across time. It specifies the conditions under which a system remains identifiable under varying load and boundary conditions.

No interpretive, psychological, or domain-specific constructs are introduced.

II. Variable Reference

The system is defined over three invariant variables: Load (L), Capacity (C), and Boundary (B). These variables are fixed at the minimal layer and are not expanded or redefined within this document.

III. Identity as Continuity

Identity is defined as the persistence of a system across time under changing load conditions.

A system remains identical to itself if structural continuity is preserved under variation in load, capacity, and boundary conditions.

Identity does not depend on system state at a single moment and is not determined by instantaneous stability or instability.

IV. Persistence Under Load

Systems are subject to varying load over time. Identity is preserved when structural continuity is maintained despite changes in load magnitude, rate, or accumulation.

Temporary instability does not terminate identity if continuity is maintained.

Collapse represents a disruption of structural continuity and may result in loss of identity under current system conditions.

V. Boundary and Identity

Boundary conditions contribute to identity by regulating exposure to load across time. Stable boundary regulation supports continuity under changing conditions.

Asymmetry or failure in boundary conditions may contribute to disruption of persistence if load exceeds the system's capacity to maintain continuity.

VI. Non-Dependence on State

Identity is not defined by stability, instability, or collapse as isolated states. It is defined by continuity across these states under structural constraints.

A system may transition between states while preserving identity, provided structural continuity is maintained.

VII. Constraint

All valid descriptions of identity reduce to structural continuity under Load, Capacity, and Boundary. Identity does not introduce additional variables and does not alter the minimal system.

VIII. Statement

Identity is defined as structural continuity across time in DHFT. All valid system descriptions preserve this condition.

IX. System Reference

The system defined in this document is specified in *Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System*, DOI: 10.5281/zenodo.19075948.

All valid system descriptions reduce to Load, Capacity, and Boundary.

No additional variables are introduced at the minimal layer.

This document is part of the DHFT technical series.

X. Copyright

This document is an original work of authorship.

No part of this publication may be reproduced, distributed, or transmitted in any form or by any means without prior written permission, except for brief quotations for academic or scholarly use.

This work defines original structural models and associated terminology within Dimensional Human Field Theory (DHFT).

© 2026 Cara Tonucci. All rights reserved.